TASK 05-50-06-200-801-A
EFFECTIVITY: ALL

1. Maximum Operating Speed Envelope Exceedance - Inspection/Check

A. General

(1) This section gives the necessary actions after an overspeed event is reported by crew or detected by flight data analysis.

(2) The exceedance of the operating speed/Mach envelope will trigger the overspeed aural warning.

(3) Do this inspection when the crew reports an overspeed aural warning or when the fight data analysis shows that the operating speed envelope was exceeded.

(4) This procedure considers V_{MO} and M_{MO} exceedance defined as follows :

- V_{MO} is the maximum operating speed.
- M_{MO} is the maximum operating Mach.

NOTE: An exceedance of any of these two parameters is considered an overspeed event.

(5) The operating speed envelope is presented in the AFM - Section 2 - Limitations, and is reproduced in Figure 601, together with the thresholds and exceedance zones defined on this procedure.

(6) Two threshold values are set to classify the severity of the overspeed event in one of the following conditions:

(a) Below or at Low-Threshold value: The low-threshold value defines an overspeed limit for which no inspection of the aircraft is necessary, if two other conditions (overspeed duration and number of occurrences per flight) are met. Low threshold limits defined as Zone A, see (Figure 601).

(b) Between Low- and High-Threshold values: This region defines a situation out of the aircraft envelope. But the operator can still restore the aircraft airworthiness, without Embraer's involvement, if no discrepancies are found during the applicable inspections. The region between Low and High-threshold is defined as Zone B, see (Figure 601).

(c) Above or at High-Threshold value: The high-threshold value defines an exceedance operating speed in a high degree of severity. In this condition, the operator still has the necessary information to restore the aircraft airworthiness, but Embraer must be informed and receive data. This region is defined as Zone C, see (Figure 601).

(7) The purpose of this section is to give the operator all information to analyze the flight data, determine the necessary inspections and restore the aircraft airworthiness. Embraer must be involved only if at least one of the situations below occurs:

(a) The highest speed recorded is above the high-threshold value. In this situation, the flight data must be sent to Embraer, but the operator is still responsible for ensuring, based on the required inspections, that the aircraft can return to service, or;

(b) During any applicable inspection, a discrepancy is found. In this situation, the operator sends to Embraer the flight data and a report of the discrepancies found. An Embraer's disposition is necessary to return the aircraft to service. In this case, Embraer will also evaluate if the findings are related to overspeed event or not. Phase III inspections will only be requested by Embraer if the findings are found to be related to event under investigation.

(8) The flight data source can be either DVDR 1, DVDR 2 or QAR. Any of them is sufficient to define if maximum operating speed envelope exceedance has occurred.

NOTE: DVDR 1 is installed in the FWD avionics compartment and DVDR 2 is installed in the aft avionics compartment.

(9) When the aircraft is equipped with the QAR, you can use this equipment as an alternative to DVDR 2.

NOTE: The QAR must operate in continuous mode to be used as alternative to DVDR 2.

(10) The operator can do the analysis of the flight operation data, but if the alternative is sending the data for an EMBRAER analysis, it is recommended that you send the QAR file.

(11) These maintenance technicians are necessary to do this task:

- 1 - Electrical/Avionics Systems
- 1 - Mechanical Systems
- 1 - Airframe (Structure)

B. References

REFERENCE	*DESIGNATION*
AMM MPP 05-50-05/600	HIGH LOAD FACTOR - INSPECTION/CHECK
AMM TASK 07-10-01-500-801-A/200	Complete Aircraft Jacking - Maintenance Practices
AMM TASK 12-11-01-650-802-A/300	Fuel Tank - Pressure Defueling
AMM TASK 20-00-00-910-801-A/200	Aircraft Maintenance Safety Procedures - Standard Procedures
AMM TASK 27-10-00-710-801-A/500	Aileron Control System - Operational Test
AMM TASK 27-20-00-710-801-A/500	Rudder Control System - Operational Test
AMM TASK 27-30-00-710-801-A/500	Elevator Control System - Operational Test
AMM TASK 27-40-00-710-801-A/500	Horizontal Stabilizer System - Operational Test
AMM TASK 27-50-00-710-801-A/500	Flap Control System - Operational Test
AMM TASK 27-60-00-710-801-A/500	Spoiler System - Operational Test
AMM TASK 27-80-00-710-801-A/500	Slat Control System - Operational Test
AMM TASK 32-12-00-210-801-A/600	Main Landing Gear Door Linkage - General Visual Inspection
AMM TASK 32-12-00-220-801-A/600	Main-Landing-Gear Doors - Detailed Dimension Inspection
AMM TASK 32-22-01-200-801-A/600	Nose-Landing-Gear Forward-Door Rods Linkage - Inspection

(Continued)

REFERENCE	DESIGNATION
AMM TASK 32-22-01-220-801-A/600	Nose-Landing-Gear Forward-Door - Detailed Dimensional Inspection
AMM TASK 51-50-00-800-801-B/600	Aircraft Alignment Check - Inspection/Check

C. Overspeed Inspection - Exceedance Reported by Crew

SUBTASK 210-005-A

(1) The necessary actions after an overspeed event are given in the flowchart shown in Figure 602.

(2) When the crew reports an overspeed aural warning, the aircraft can return to service without any other action if the 3 conditions that follow are met:

(a) Condition 1: The speed stayed below 330 KIAS, or Mach stayed below M 0.83 throughout the flight;

(b) Condition 2: The overspeed event did not last more than 20 SEC.

(c) Condition 3: There was only two overspeed event in the same flight.

(3) If the crew cannot assure that all the three conditions above were respected, it is necessary to do the fight data download and data must be analyzed.

NOTE: If it is possible to do the flight data analysis before next flight, use the Figure 603.

(4) *EFFECTIVITY: ON AIRCRAFT WITH HONEYWELL DVDR*

If it is not possible to do the fight data analysis before next flight, do Phase I inspection. If no defect is detected, a deferrable period of 10 FC is permitted.

NOTE: The flight data download must be done within a maximum of 20 FH. If not, the data can be missed due to the recording time of the unit.

(5) *EFFECTIVITY: 190:00378-00429 00434-00434 00449-00449 00458-00458 00471-00491 00498-00498 00510-00686 00693-00697 00699-99999 OR 190:00163-00242 00282-00282 00329-00329 00450-00450 00460-00460 00493-00495*

If it is not possible to do the fight data analysis before next flight, do Phase I inspection. If no defect is detected, a deferrable period of 10 FC is permitted.

NOTE: The flight data download must be done within a maximum of 120 FH. If not, the data can be missed due to the recording time of the unit.

(6) *EFFECTIVITY: 190:00271-00271 00330-00330 00688-00692 00698-00698*

If it is not possible to do the fight data analysis before next flight, do Phase I inspection. If no defect is detected, a deferrable period of 10 FC is permitted.

NOTE: The flight data download must be done within a maximum of 120 FH. If not, the data can be missed due to the recording time of the unit.

(7) In the deferrable period, the fight data must be analyzed and compared with the threshold values and conditions shown on Table 601. This will give you the necessary actions.

(8) Whenever an overspeed event is investigated by flight data analysis, a new condition must be observed, as shown below:

(a) Condition 4: The normal acceleration must stay below 2G's.

NOTE: If the flight data analysis is not available, do the phase-II inspection during the 10 FC deferrable validity.

(9) For the event assessment, the parameters below must be selected from the fight data:

- Time (GMT)
- Calibrated Airspeed (Airspeed-Pil and Airspeed-CoP)
- Mach
- Normal acceleration (NormAccel)
- Pressure Altitude

NOTE: The parameters must be selected from a fight data source with a sample rate of 8 sps (samples per second).

D. Overspeed Inspection - Exceedance Detected by Flight Data Analysis

SUBTASK 210-010-A

(1) The necessary actions after an overspeed event detected by flight data analysis are given in the flowchart Figure 603.

(2) The same condition applicable to an overspeed reported by crew also applies and are shown on Table 601.

NOTE: This condition is also observed when performing flight data analysis after a overspeed report by crew.

E. Exceedance Detection

SUBTASK 210-006-A

(1) In the fight data, identify the time interval where the overspeed occurred.

(2) Select the highest measured value for Mach and KCAS. Compare both values with the threshold values given in Table 601:

Table 601 - THRESHOLD VALUES

PARAMETER	LOW THRESHOLD ($T_{L)}$	HIGH THRESHOLD ($T_{H)}$
KCAS	330 kts	350 kts
Mach	0.83	0.85

(3) Based on the position of the highest sample recorded, the overspeed event will be placed in one of these 3 regions (refer to Figure 601):

1. Zone A - Highest sample Below or at T_{low} (≤ 330 KCAS or M 0.83)

1. Zone B - Highest sample between T_{low} and T_{high} (330 to 350 KCAS or M 0.83 to 0.85)

1. Zone C - Above or at T_{high} (≥ 350 KCASor M 0.85)

(4) Take note of the duration of the overspeed event, as this will be used to classify its severity.

(5) Analyse the data of the entire flight and count the number of times the speed was out of the envelope. This parameter will be also used to classify the severity of the event.

(6) Refer to Table 601 to classify the event in the applicable zone and to find the necessary actions, for each zone:

Table 602 - Zones and Required Actions

VALUE	REQUIRED ACTIONS
Zone A	No actions required if conditions 1 to 3 are observed.[1]
	If condition 2 or 3 are not observed, consider the event as being a Zone B event.[1]
Zone B	Phase I inspection before the next flight.
	Phase II following phase I inspection, if any discrepancy detected during phase I.[2]
	Phase II inspection during 10 FC deferrable if no discrepancy is found on phase I inspection.
	Phase III inspection required only if discrepancies detected during phase I or II.[2] [3]
Zone C	Phase I inspection before the next flight.
	Phase II and III inspections following phase I, if any discrepancy detected during phase I inspection.[2]
	Phase II and III inspections during the 10 FC deferrable if no discrepancy found on phase I inspection.[4]

[1] See the conditions 1 to 3 at SUBTASK 210-005-A item (2).
[2] Send flight data and data of findings to Embraer, if discrepancies are found during phase I or II inspections.
[3] Phase III will be required by Embraer if the findings are determined to be related to the overspeed event.
[4] Phase III mandatory for events on Zone C.

(7) In the same time interval where the speed was out of the envelope, check the highest value for Normal Acceleration. If the highest value exceeds 2 g, do AMM MPP 05-50-05/600.

F. Job Set-Up

SUBTASK 940-004-A

WARNING: **MAKE SURE THAT THE AIRCRAFT IS IN A SAFE CONDITION BEFORE YOU DO THE MAINTENANCE PROCEDURES. THIS IS TO PREVENT INJURY TO PERSONS AND/OR DAMAGE TO THE EQUIPMENT.**

(1) Do the procedure to make the aircraft safe for maintenance (AMM TASK 20-00-00-910-801-A/200).

G. Phase-I Inspection

SUBTASK 210-007-A

(1) Do an external visual inspection of the entire aircraft for missing components, such as doors, access panels and fairings.

(2) Do an inspection of the aircraft doors and access panels for distortion, broken or missing latches, and missing fasteners.

(3) Do a visual inspection of the aileron assembly for general condition.

(4) Do a visual inspection of the rudder assembly for general condition.

(5) Do a visual inspection of the elevator assembly for general condition.

(6) Do a visual inspection of the inboard and outboard flaps assemblies for general condition.

(7) Do a visual inspection of the slat assembly for general condition.

(8) Do a visual inspection of the spoilers assemblies for general condition.

(9) Make sure that the aileron, rudder, elevator, flaps, slats and spoilers move freely.

H. Phase-II Inspection

SUBTASK 210-008-A

(1) Fuselage Inspection

(a) Do an external visual inspection of the fuselage skin (bottom, side and crown) for distortion, buckling and pulled or missing rivets.

(b) Carefully examine the areas comprised between frames 43 and 58 and between frames 90 and 100.

NOTE: If you find any discrepancy during the external inspection, examine the internal structure for damage. In this case, look for damage on the skin, ribs, frames and attachments.

(c) Examine the internal structure of the rear fuselage, between frames 90 and 100. Look for distortion, buckling and pulled or missing rivets, with particular attention to:

- aft pressure bulkhead;
- machined frames;
- pitch trim actuator and its frames attachments;
- hinges and fittings.

(d) Examine the area adjacent to the fuselage bulkheads in the attachment region of wing spars I, II and III, and empennage. If you find any external damage, examine all the internal structure in the damaged areas.

(e) Examine the seal fairing of wing-to-fuselage fairing side panel.

(f) Inspect the windshield wiper for general condition.

(2) Wing Inspection

(a) Externally examine the upper and lower wing surfaces. Look for general misalignment, missing or pulled fasteners. If you find any discrepancy, examine the internal structure for damage.

(b) Examine the external surface of the trailing edge for buckles. If you find any discrepancy, examine the internal structure for damage.

(c) Examine the winglet surfaces for distortion and missing or pulled fasteners.

(d) Examine the external surface of the wing-to-fuselage fairing. Look for signs of distortion and missing or pulled fasteners.

(e) Do a inspection of the wing stub spar III.

(f) Inspect the MLG bay for signs of fuel leaks.

(3) Empennage Inspection

(a) Do an inspection of the horizontal stabilizer lower and upper forward and aft splice-plates.

(b) Do an inspection of the horizontal stabilizer upper and lower skin.

(c) Do an inspection of the vertical stabilizer-to-rear fuselage attachment.

(d) Deflect the rudder and elevator and examine the external spars. Look for distortion, buckling, missing or pulled fasteners. If you find any discrepancy, examine the internal structure for damage.

(e) Examine the dorsal fin external surface for signs of buckling.

(f) Check the horizontal and vertical stabilizers tip fairings for cracks and missing fasteners.

(g) For the elevator and rudder, check the hinges for integrity.

(h) Examine the aerodynamic seals for integrity.

(i) Examine the access panels for general condition and missing fasteners.

(4) Flight Controls Inspection

(a) Do an inspection of the elevator upper and lower skins.

(b) Do an inspection of the elevator hinges, fittings and actuator.

(c) Do an inspection of the rudder skin.

(d) Do an inspection of the rudder hinges, fittings and actuators.

(e) Do an inspection of the aileron skin.

(f) Do an inspection of the aileron hinges, fittings and actuator.

(g) Do a check of all the surfaces for correct alignment at neutral position.

(h) Do an inspection of the rudder leading edge for signs of chafing with the vertical empennage fairing.

(i) Examine the aerodynamic seals of all flight-control surfaces for integrity.

(j) Examine the access panels for general condition and missing fasteners.

(5) NOTE: The aircraft do not need to do MLG and NLG doors adjustments at phase II inspection if the exceedance time was above 20 seconds but the speed limit was below Low TV (330 Knts).

MLG and NLG Doors Adjustments;

(a) Do a check of the doors for general misalignment (and);

(b) Do a check of the door mechanism for bowed or distorted rods (AMM TASK 32-12-00-210-801-A/600 and AMM TASK 32-22-01-200-801-A/600);

(c) Do a check of the rigging of the assembly (AMM TASK 32-12-00-220-801-A/600 and AMM TASK 32-22-01-220-801-A/600).

(6) Pylons and Nacelles

(a) Do an inspection of the pylons access panels for buckling, cracks, missing or pulled fasteners.

(b) Do a check of the aerodynamic seals for general condition.

(c) Do an inspection of the nacelles for signs of fuel and other fluid leaks.

I. Phase-III Inspection

SUBTASK 210-009-A

(1) Do the operational checks of the:

- Aileron (AMM TASK 27-10-00-710-801-A/500).
- Inboard and outboard flaps (AMM TASK 27-50-00-710-801-A/500).
- Slats (AMM TASK 27-80-00-710-801-A/500).
- Spoilers (AMM TASK 27-60-00-710-801-A/500).
- Rudder (AMM TASK 27-20-00-710-801-A/500).
- Horizontal stabilizer (AMM TASK 27-40-00-710-801-A/500);
- Elevator (AMM TASK 27-30-00-710-801-A/500).

(2) Defuel the aircraft (AMM TASK 12-11-01-650-802-A/300).

(3) Jack the aircraft (AMM TASK 07-10-01-500-801-A/200).

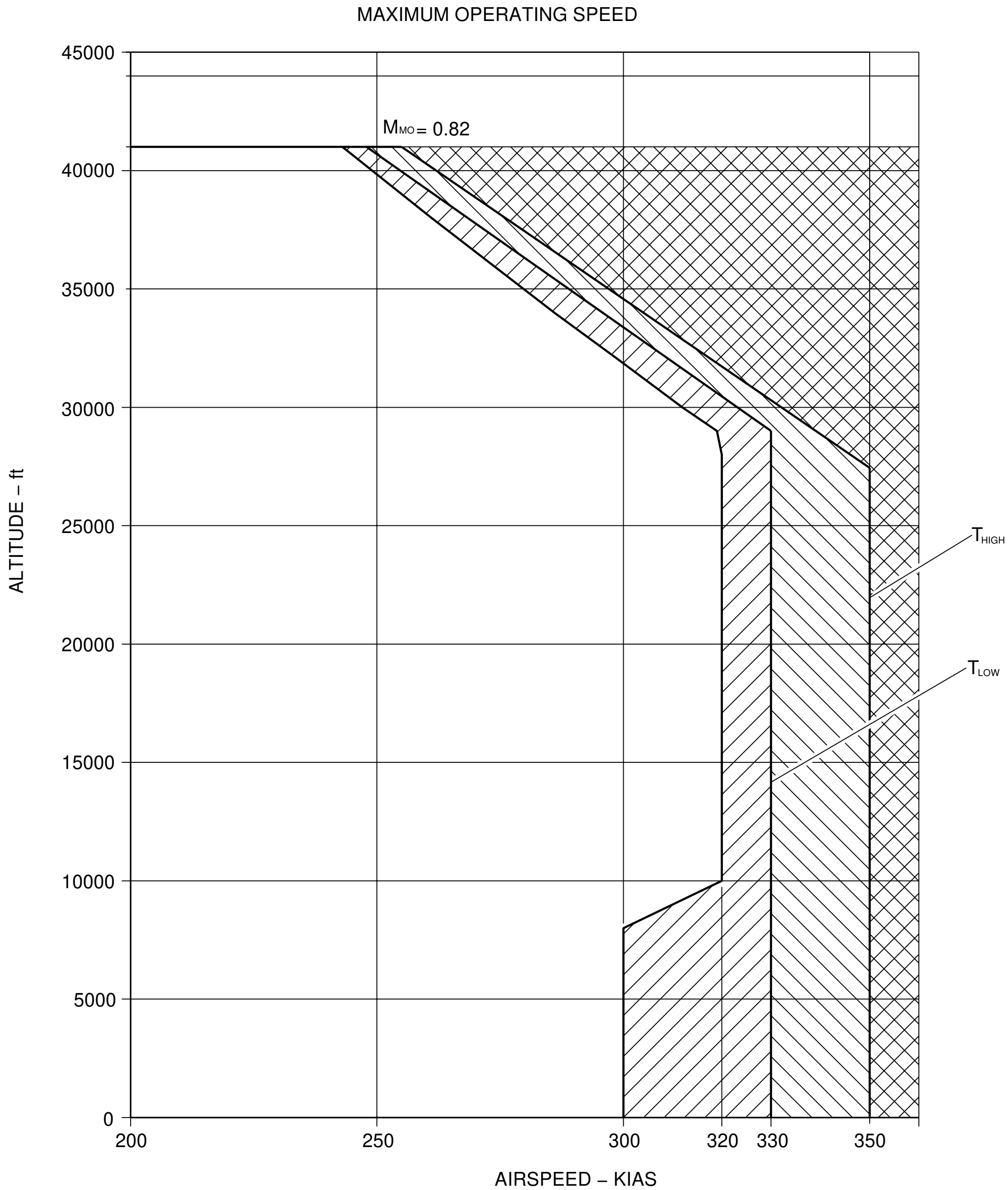
(4) Do a check of the aircraft alignment. Refer to AMM TASK 51-50-00-800-801-B/600.

J. Job Close-Up

SUBTASK 940-005-A

(1) Put the aircraft back to its initial condition (AMM TASK 20-00-00-910-801-A/200).

EFFECTIVITY: ALL
Operational Speed Envelope
Figure 601



LEGEND:

ZONE A – BETWEEN OPERATIONAL ENVELOPE AND T_{LOW}

ZONE B – BETWEEN T_{LOW} AND T_{HIGH}

ZONE C – ABOVE T_{HIGH}

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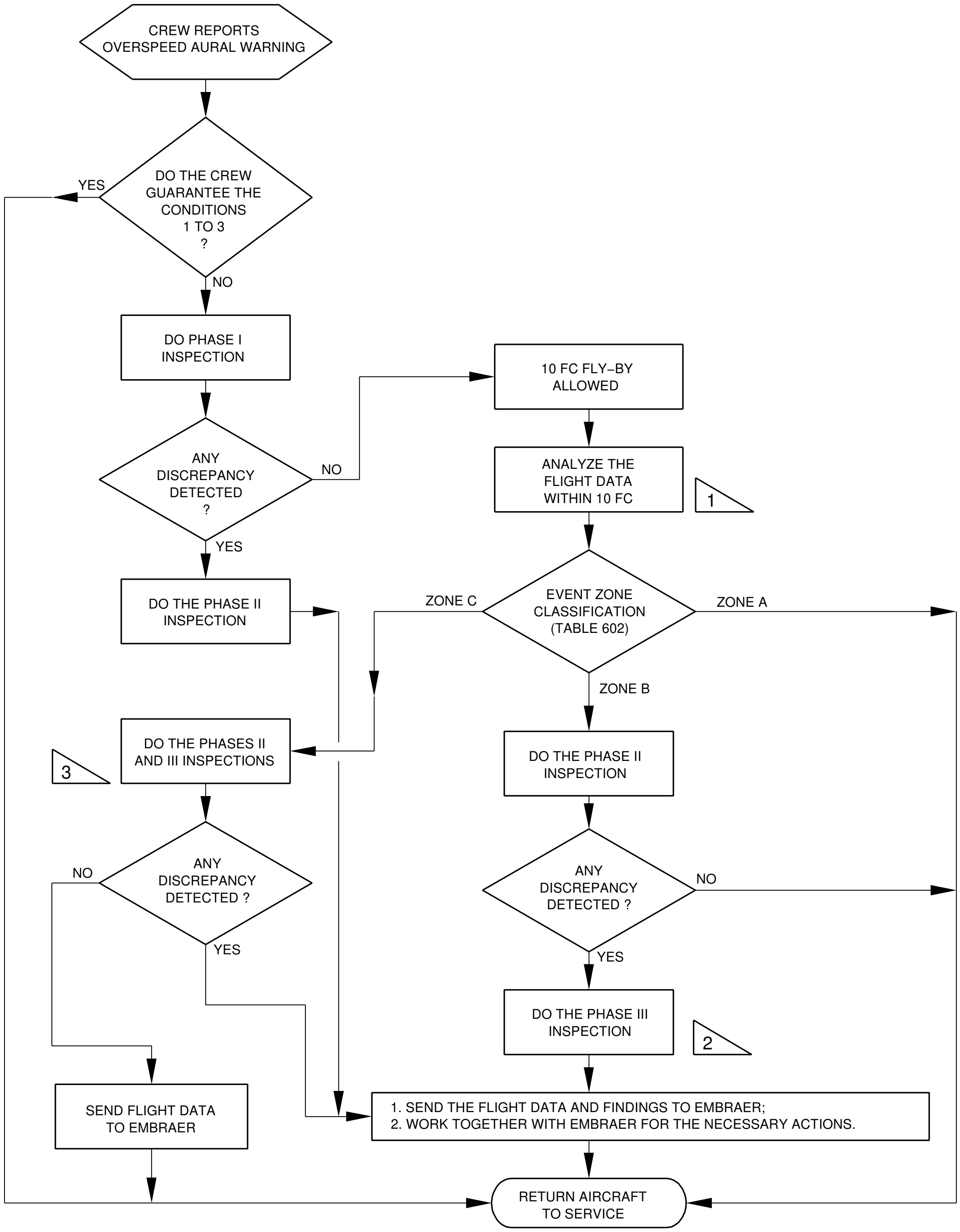
EFFECTIVITY: ON AIRCRAFT WITH HONEYWELL DVDR
Overspeed Detection - Crew Report
Figure 602 - Sheet 1



1 DO THE FLIGHT DATA DOWNLOAD IN A MAXIMUM OF 20 FH, OTHERWISE YOU CAN MISS THE DATA BECAUSE OF THE LIMITED RECORDING TIME OF THE UNIT.

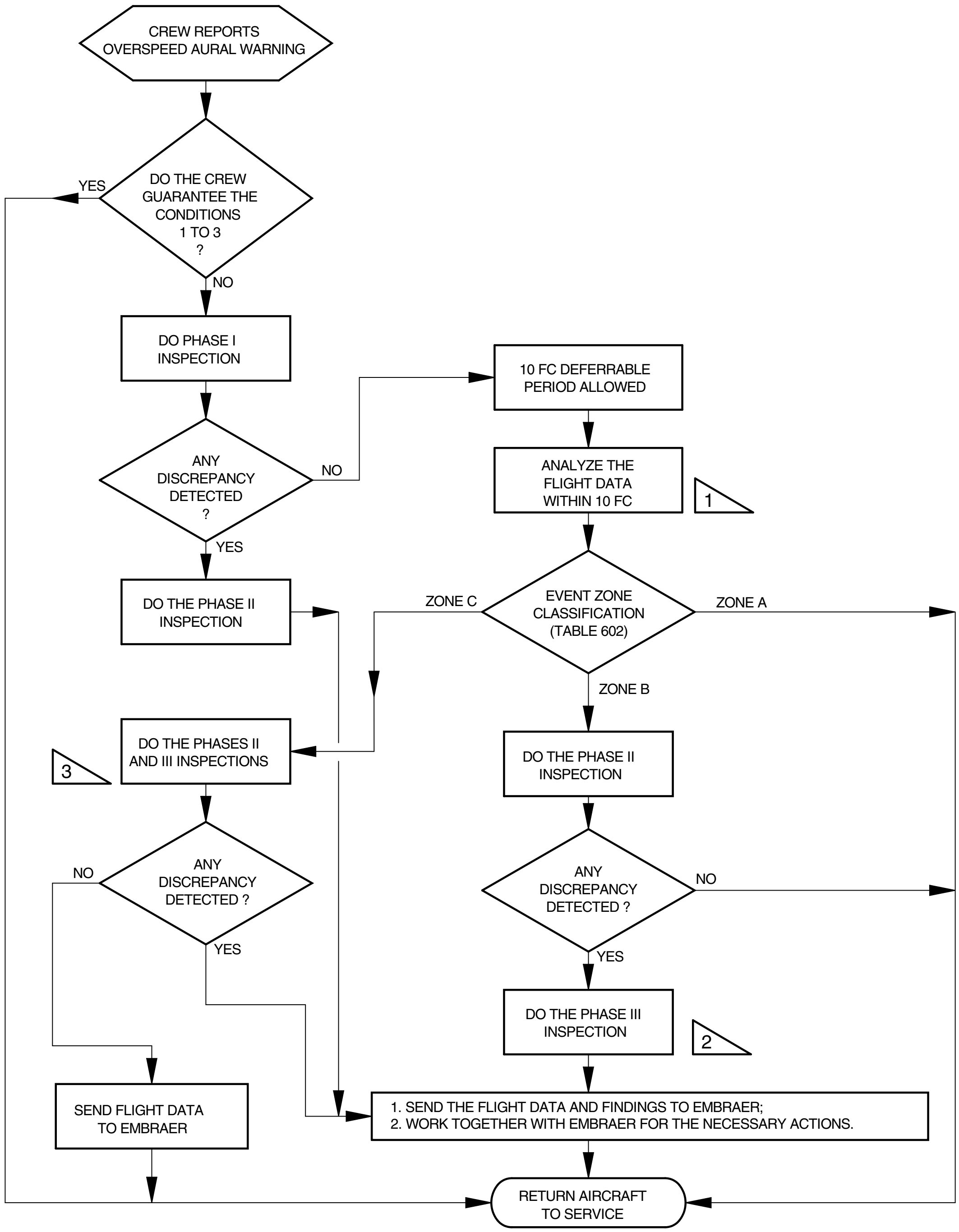
2 EMBRAER WILL EVALUATE THE FINDINGS AND DEFINE IF RELATED TO THE OVERSPEED EVENT OR NOT AND INDICATE THE NEED FOR PHASE III

3 PHASE III INSPECTION IS ALWAYS REQUIRED FOR EVENTS ON ZONE C

EM170MPP050098J.DGN

EFFECTIVITY: 190:00378-00429 00434-00434 00449-00449 00458-00458 00471-00491 00498-00498 00510-00686 00693-00697 00699-99999 OR 190:00163-00242 00282-00282 00329-00329 00450-00450 00460-00460 00493-00495

Overspeed Detection - Crew Report
Figure 602 - Sheet 2



1 DO THE FLIGHT DATA DOWNLOAD IN A MAXIMUM OF 120 FH, OTHERWISE YOU CAN MISS THE DATA BECAUSE OF THE LIMITED RECORDING TIME OF THE UNIT.

2 EMBRAER WILL EVALUATE THE FINDINGS AND DEFINE IF RELATED TO THE OVERSPEED EVENT OR NOT AND INDICATE THE NEED FOR PHASE III

3 PHASE III INSPECTION IS ALWAYS REQUIRED FOR EVENTS ON ZONE C

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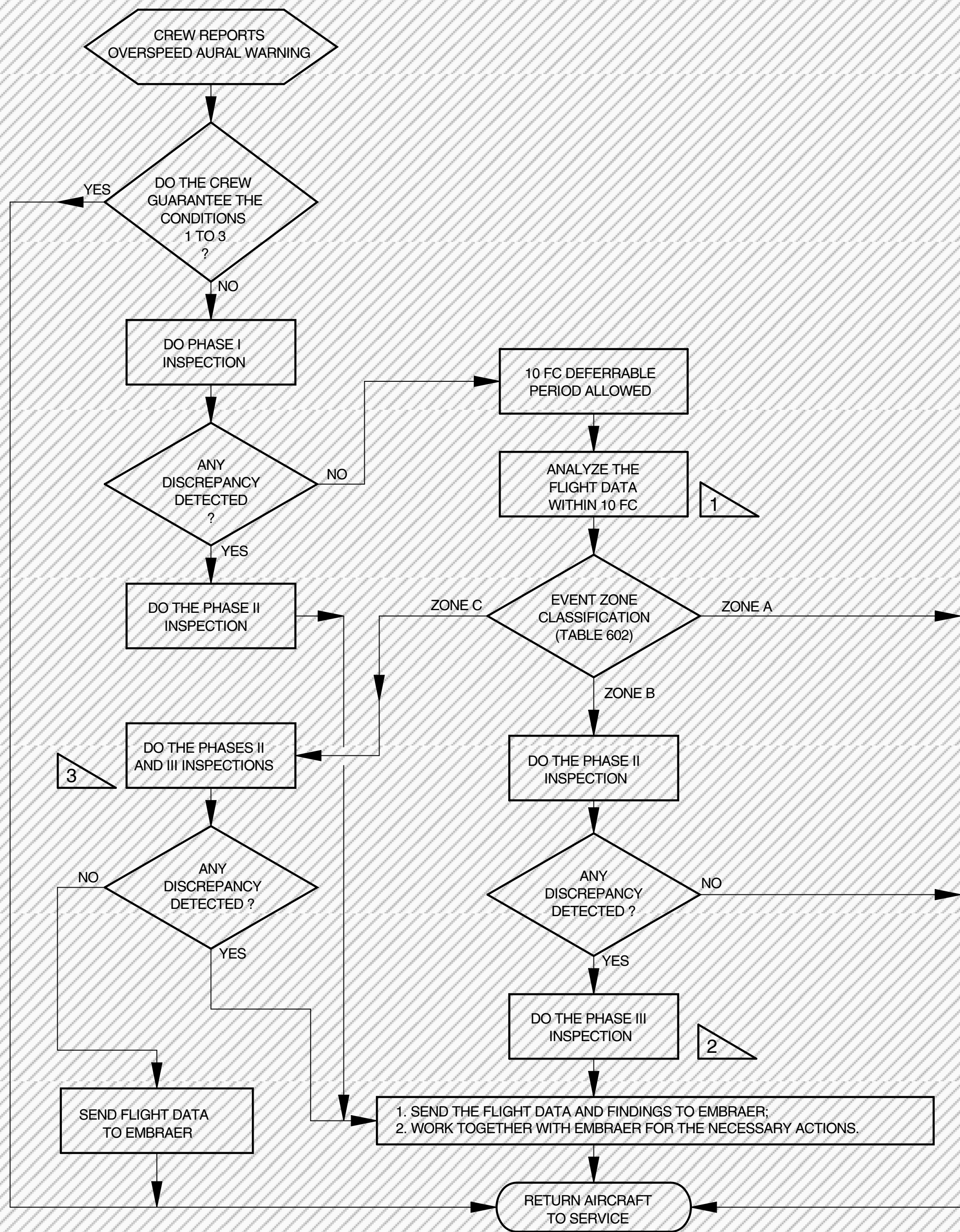
EFFECTIVITY: 190:00271-00271 00330-00330 00688-00692 00698-00698

Overspeed Detection - Crew Report

Figure 602 - Sheet 3



1 DO THE FLIGHT DATA DOWNLOAD IN A MAXIMUM OF 120 FH, OTHERWISE YOU CAN MISS THE DATA BECAUSE OF THE LIMITED RECORDING TIME OF THE UNIT.

2 EMBRAER WILL EVALUATE THE FINDINGS AND DEFINE IF RELATED TO THE OVERSPEED EVENT OR NOT AND INDICATE THE NEED FOR PHASE III

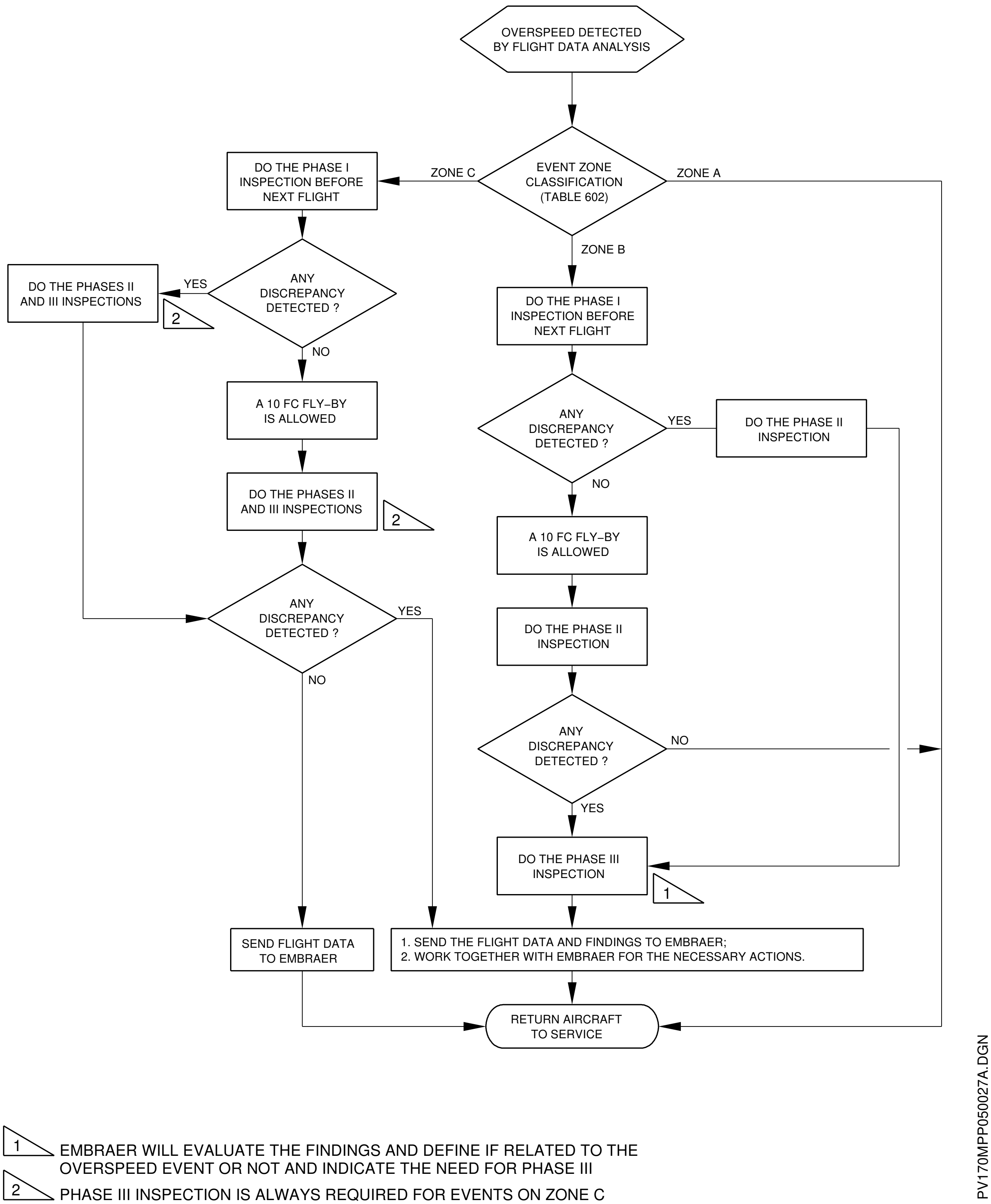
3 PHASE III INSPECTION IS ALWAYS REQUIRED FOR EVENTS ON ZONE C

EM170MPP059100B.IDR

EFFECTIVITY: ALL

Overspeed Detection - Detected by flight data analysis

Figure 603



1 EMBRAER WILL EVALUATE THE FINDINGS AND DEFINE IF RELATED TO THE OVERSPEED EVENT OR NOT AND INDICATE THE NEED FOR PHASE III

2 PHASE III INSPECTION IS ALWAYS REQUIRED FOR EVENTS ON ZONE C